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# Design of regional production networks for second generation synthetic bio-fuel – A case study in Northern Germany

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ABSTRACT

In the medium-term, second generation synthetic bio-diesel will make an important contribution to sustainable mobility. However, attributed to political, technical, and market related uncertainties, it is still not clear which interest groups will invest in production capacities and which technologies will be used. Hence, a multi-period MIP-model is presented for integrated location, capacity and technology planning for the design of production networks for second generation synthetic bio-diesel. The approach is applied to the region of Niedersachsen, Germany. Principle network configurations are developed for this region considering different scenarios and different risk attitudes of interest groups. As results of the investigation, recommendations are drawn regarding advantageous plant concepts, as well as strategies for the capacity installation. Finally, recommendations for political decision makers as well as for potential investors are deduced.



## 1. Introduction

Climate change, shortage of the finite oil resources, and political developments in oil-exporting countries require new means of energy supply. The traffic sector contributes with 48% to global oil demand (US Energy Information Administration, 2009), and 20–30% of all anthropogenic $CO_2$-emissions of developed countries arise from this sector (UNFCCC, 2007). The use of bio-fuels is a promising option to meet future energy supply and to reduce $CO_2$-emissions. However, presently used first generation bio-fuels are criticized, since increasing food prices are to a certain extend traced back to bio-fuels production and ecological damages result from the exploitation of arable land for the cultivation of biomass. These negative effects can be avoided by the use of second generation bio-fuels, which can be produced from residual materials, e.g. straw and logging residues. Since the complete lignocellulosic material is processed, higher area yields are possible (Berndes et al., 2010).

Due to the very early development stage of production technologies for second generation ethanol and a high demand for diesel in Europe, the focus of this paper will be on synthetic bio-diesel. The production of synthetic bio-diesel is characterized by various uncertainties regarding future availability of raw materials, progress in production technologies, future diesel demand, and political conditions. Due to these uncertainties, it is still unknown which interest groups (e.g. petroleum industry, automotive industry, agricultural consortia) will invest in production capacities. Therefore, the aim of this paper is to analyze the planning problem of selecting and locating production capacities for synthetic bio-diesel production in the face of different risk attitudes of possible decision makers. We develop a problem specific planning approach and apply it to a case study. This allows us to evaluate under which conditions and for which investors the production of synthetic bio-diesel is beneficial.

The paper is organized as follows. In section 2, we derive requirements that must be met by the planning approach and analyse existing planning models with regard to these requirements. In section 3, we present a specific MIP-model comprising all aspects of production planning for second generation bio-diesel. We extend this model into a scenario based planning approach that takes the different risk settings of potential investors into account. In section 4, we present data, scenarios and resulting network configurations for a production network for bio-diesel in northern Germany. Based on these results, recommendations for investors and political decision makers are derived.

## 2. Planning problem

In the following, characteristics of production networks for second generation bio-diesel and decisions to be taken in the course of the implementation of these networks are described. This allows the derivation of requirements that must be fulfilled by a planning

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approach for the design of such networks. Existing models are then analysed regarding these requirements.

### 2.1. Characteristics of the planning problem

A potential bio-diesel production network can be divided into three main parts: biomass generation, bio-diesel production, and bio-diesel demand.

First, the biomass is gathered in agricultural regions. It is expected that fast-growing lumber, logging remains and agricultural waste will be used for production of bio-diesel. The different biomasses differ strongly in terms of energy and water content. Future availability, quality and prices of biomass are uncertain because of unknown developments in plant breed, climatic conditions, and intensity of land use.

Afterwards, the biomass is transported to production sites. Within the production part of the network, the transformation of biomass into bio-diesel takes place. At present, different production technologies for synthetic bio-fuels are under development e.g. in Austria (Guessing, 2011; Hofbauer et al., 2006), the US (US Department of Energy, 1998), the Netherlands (Hamelinck and Faaij, 2006), and in Germany (Henrich et al., 2007; Choren, 2011; Radig et al., 2006; Maly and Claußen, 2005). These technologies all have the main process steps in common, but differ with regard to the kind and sequence of unit operations and the overall network concept. Fig. 2.1 shows the general production process and two specific plant concepts: the decentralized plant concept of the Forschungszentrum Karlsruhe (FZK) (Henrich et al., 2007) and the centralized plant concept of Choren (2011).

In the concept of Choren, gasification of dried biomass and subsequent synthesis and refining all take place within a centralized plant. Thus, economies of scale can be achieved. In the decentralized plant concept of FZK, bulky biomass is first transformed into an energy dense and transportable intermediate, the so-called slurry, applying a pyrolysis process close to the biomass provision sites. Gasification of slurry, subsequent synthesis and refining takes place in a second step. Hence, fewer transports of biomass are required, but economies of scale in production can only be achieved within the second production step. Different gasification and synthesis processes can be used (e.g. Fischer–Tropsch-Synthesis or Methanol Synthesis) for centralized as well as for decentralized technologies. Altogether, process design options and estimations on future process yields, achievable economies of scale, and investments are very uncertain, since large industrial realizations of the different technical options do not yet exist.

The last part of the network is characterized by demand for bio-diesel. This demand depends on the effective ratio of bio-diesel on overall diesel demand. A minimum ratio of bio-diesel is set by mandatory blending rates. For instance, the latest EU directive 2009/28/EC requires a 10% substitution of fossil fuel through bio-fuel by the year 2020. There is no maximum ratio, i.e. there are no technical reasons that limit the use of synthetic bio-diesel. Thus, higher ratios than mandated by legislation will depend on the price of bio-diesel and fossil diesel. Since these prices are uncertain, effective ratio of bio-diesel on overall diesel demand and thus total bio-diesel demand is hard to predict and very uncertain. Additionally, a dynamic set-up of production capacities is required as mandatory blending rates increase over time.

A contemporary and extensive production of synthetic bio-fuels requires a targeted political support. However, it is currently still unclear for political decision makers, which interest group will invest. The petroleum industry, the automotive industry, as well as agricultural consortia have fundamental interest in production and distribution of synthetic bio-diesel. However, these interest groups are confronted with specific legal and financial conditions and thus show different risk attitudes towards high investment costs for production technologies. Since it is also open, which technologies will be used, it is necessary to carry out a general assessment of the installation of production technologies and capacities considering the different risk attitudes of potential investors for the overall production network.

### 2.2. Model requirements

Different planning problems can be identified. On the one hand, the diesel production has to be designed. To do this, plant concepts,

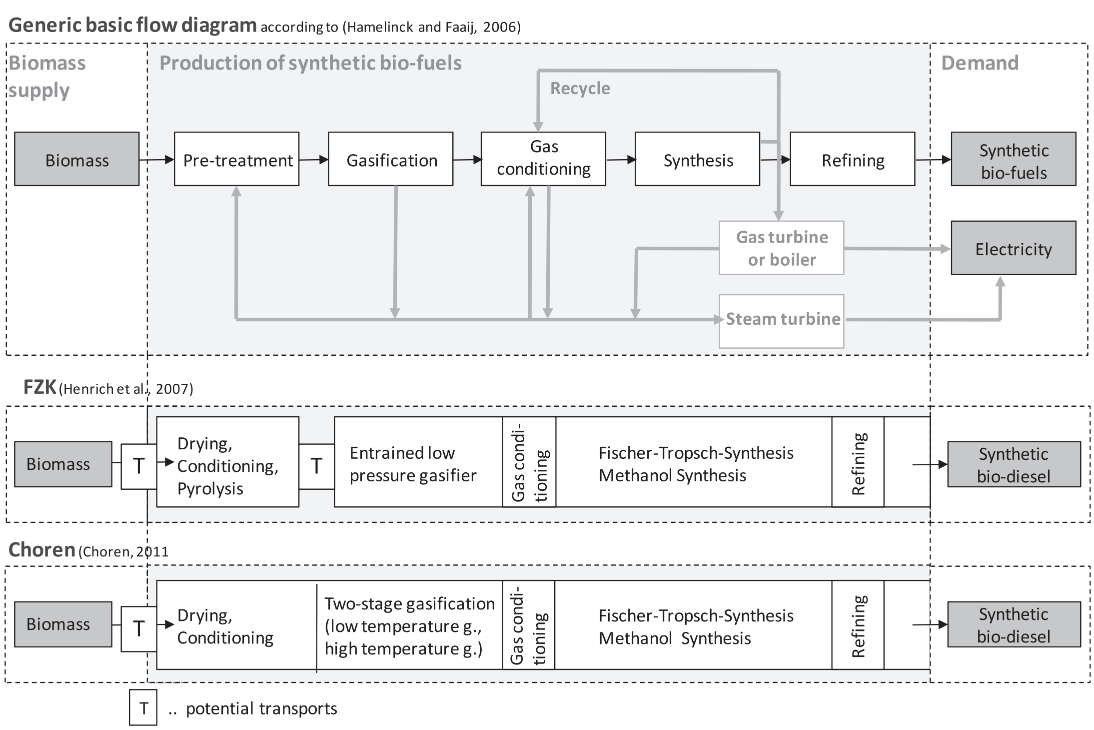



**Fig. 2.1.** Production concepts for synthetic bio-diesel.

capacities, and technologies have to be selected and installation periods and locations have to be fixed. On the other hand, transformation and allocation of the material flows have to be performed within the network structures.

We will thus develop an approach for integrated location, technology and capacity planning. Certain requirements must be met, which can be summarized as follows:

- Modelling of the material transformation in chemical and physical processes of different capacities for centralized as well as decentralized plant concepts.
- Consideration of known trends in general conditions, such as increasing mandatory quotas for bio-diesel in the course of time.
- Consideration of uncertainties regarding future developments as well as different risk attitudes of potential investors.
- Usage of a profit-oriented approach regarding discounted revenues and costs over the complete planning horizon in order to evaluate the profitability of the network.

Average material flows and annual planning periods are used because of the strategic character of the planning problem, and since the production processes operate consistently. Seasonal effects and inventory costs have a negligible influence for the strategic planning problem studied in this paper. Budget constraints would have to be taken into account if network planning would be carried out for an individual investor. Since the aim is to evaluate under which conditions and for which investors the production of synthetic bio-diesel is beneficial, budget constraints are neglected.

### 2.3. Review of existing planning models

The deduced requirements are cross-checked with existing models in facility location planning, respectively network design. In doing so, we reveal that the required characteristics are individually covered in existing modelling approaches, but can not be found within one single planning model. Thus, there is a need to develop a problem specific model for the design of production networks for synthetic bio-diesel.

Reviews on existing models in the field of facility location planning can be found in Klibi et al. (2010), Melo et al. (2009), Snyder (2006), ReVelle and Eiselt (2005), and Klose and Drexl (2005) for example. In these reviews, the authors classify the planning models e.g. with regard to network structure (number of transportation stages), planning horizon (single or multi-period), number of materials, consideration of capacity limits, as well as with regard to uncertainty (deterministic, stochastic, and scenario-based).

In contrast to pure distribution networks, material transformations and accordingly multiple materials are considered in production networks. Within the manufacturing industry, products are assembled from components, as modelled by Sabri and Beamon (2000); Yan et al. (2003). The assembly can be described by operating charts and parts lists. Process industry, on the contrary, is characterized by chemical and physical material transformations which are described by means of transformation coefficients, e.g. in Hübner (2007), Kallrath (2002). These transformation coefficients can either be derived empirically from existing processes or from technical simulation studies.

Previous models for location planning of bio-fuel production plants were developed by Leduc et al. (2010), Kerdoncuff (2008) and Leduc et al. (2008). These models take into account one specific plant concept exclusively (centralized or decentralized). That way, a rigid network structure is constituted. Since one cannot foresee how production technologies will develop in the long-term, flexible network structures are required to be able to react to modifications of the plant concepts. Such flexible models exist for other applications, e.g. in (Wollenweber, 2008; Melo et al., 2006; Martel, 2005; Spengler et al., 1997).

Modifications of model data in the course of time can be taken into account using multi-period models. Due to the long-term orientation of network design decisions, a cash flow oriented assessment is appropriate considering discounting effects (Kallrath, 2002; Fleischmann et al., 2006). Nevertheless some authors disregard these effects, such as (Melo et al., 2006; Shulman, 1991).

The long-term developments are accompanied by uncertainties that must be taken into account. In principle, literature distinguishes between reactive and proactive approaches (Klibi et al., 2010; Snyder, 2006; Mulvey et al., 1995). Planning models assuming a certain database are used in reactive approaches. The effects of modifications of data on the objective are analyzed subsequently. Accordingly, effects on the objective can be shown, but cannot be influenced during the planning process. Proactive approaches, on the contrary, explicitly consider uncertainties in planning models (Scholl, 2001; Mulvey et al., 1995).

In this context, stochastic linear programming is frequently used (Kall and Mayer, 2005; Snyder, 2006). However, risk attitudes of decision makers are not considered by stochastic linear programming. To overcome this disadvantage, Mulvey et al. (1995) and Kouvelis and Yu (1997) established different approaches of robust optimization. In robust optimization, uncertainties are described through different scenarios. Kouvelis and Yu (1997) assume in their discrete robust planning approach extreme risk averse decision makers, as they optimize the worst objective value (maxmin). In contrast, Mulvey et al. (1995) use the scenarios' occurrence probabilities. Chen et al. (2006) for example combine these approaches by using the scenarios' occurrence probabilities to determine a set of different worse case scenarios within a regret model. The use of occurrence probabilities allows not only for the modeling of expected values, but also for the modeling of higher moments of objective functions. Thus, statements regarding the underlying risk of the detected solutions can be made. The solutions are determined by two kinds of decision variables. In analogy to a two stage formulation of stochastic programs, Mulvey et al. (1995) introduce design and control variables. Design variables describe decisions, which are identical for all scenarios. In network design, those are typically the location, capacity, and technology selection. Control variables determine adjustments for each scenario individually, like the planning of material flows within the network structures (Klibi et al., 2010; Lee et al., 2010).

In most cases, an unrestricted application of robust optimization to network design is impossible. Generally, reliable data from the past does not exist for such long-term and unique decisions as location planning. Occurrence probabilities of scenarios thus cannot be derived and the planning situation is hence characterized by uncertainty according to Rosenhead et al. (1972). In this context, some authors, like Serra and Marianov (1998), use worst-case formulations according to Kouvelis and Yu (1997).

The planning situation considered in this paper is novel, due to the new technologies and the general conditions. Thus, no distribution functions are available for the uncertain parameters. Hence, a planning approach is required that considers uncertainties through scenarios. If not only very risk averse decision makers are to be represented, the scenarios' subjective occurrence probabilities have to be defined by experts.

Uncertainties may relate to both, the objective function and the constraints of a model. In case constraints are affected, it may be reasonable not to insist on their compliance. In compensation models, violations of the original constraints will accordingly be punished by penalty costs. Thus, in compensation models only feasible solutions are generated. In practice, those penalty costs may result from the acquisition of additional capacities. Another kind of relaxation is done in chance constraint models. Here, certain probabilities are defined for violating single or all constraints (Scholl, 2001). The application of chance constraint models, however, requires a large number of scenarios in order to define reasonable violation levels of constraints. As these are not available for the considered planning situation, compensation models will be used for the network design of synthetic bio-diesel.

For solving large-scale network design problems, standard solving procedures (implemented in commercial solvers) as well as specific algorithms are used. The standard solving procedures are either used to determine the optimal solution as in Salema et al. (2007); Fleischmann et al. (2006), or to determine a slightly worse solution by setting a MIP gap tolerance (Cordeau et al., 2006; Papageorgiou et al., 2001) or a limit for the computing time (Ambrosino and Scutella, 2005; Melkote and Daskin, 2001). Due to the high complexity of most network design problems, many problem specific algorithms were developed. Most of the algorithms take advantage of the special structure of the network design problem. Algorithms aiming at an optimal solution often decompose the problem into the linear flow problem and the binary problem of technology and/or location selection using Benders decomposition as in Geoffrion and Graves (1974). Santoso et al. (2005) use this basic approach for network design under risk and combine a Benders decomposition algorithm with a sample average approximation. Within heuristic algorithms, Lagrange relaxation is commonly used. Often, the connection between the production and distribution stages (Hinojosa et al., 2008) and the capacity restrictions (Klose, 2000) are relaxed. Other solution procedures are based on meta-heuristics (Jayaraman et al., 2003; Lin et al., 2006) or linear relaxations (Gunnarsson et al., 2004). A heuristic solution procedure for network design under risk combining a sample average approximation and a dual decomposition is for example presented by Schütz et al. (2009).

As can be seen, requirements for planning of production networks of synthetic bio-diesel are regarded by existing models. However, there is no single planning model in the field of general facility location planning/network design that accommodates all necessary characteristics sufficiently. Planning approaches for design of bio-fuel production networks provide not enough flexibility to account for centralized and decentralized plant concepts simultaneously. Therefore, a case specific planning model is presented in the following.

## 3. Model formulation

Within a first step, a basic multi-period, multi-stage facility location model is formulated that allows for reactions to changing conditions and is flexible to integrate centralized and decentralized plant concepts. The material transformation in production facilities is described by transformation coefficients. In a second step, the basic model is extended into a scenario based planning approach applying different alternative objective functions. These objective functions represent different risk attitudes of decision makers. As it is uncertain which developments will occur in the future, the models consider the scenarios simultaneously.

### 3.1. Basic model

The system encompassing biomass supply, transportation, production and demand can be adequately modelled as network (cp. Fig. 3.1). Several types of biomass ($B$) are provided at the network's sources ($Q$) and are transported to the production locations ($P$). Here, the transformation of biomass into synthetic bio-diesel takes place. In centralized plant concepts, production modules of eligible capacities $\left(m \in M_{in}^{biomass} \cap M_{out}^{biodiesel}\right)$ directly produce bio-diesel. In decentralized concepts, production is carried out in two different module types: the intermediate slurry is produced in a first module type $\left(g \in M_{out}^{slurry}\right)$, and bio-diesel is generated in a synthesis module $\left(m \in M_{in}^{slurry}\right)$ in a subsequent process. The synthetic bio-diesel is transported to the sinks of the network ($S$) in order to satisfy demand. The planning horizon is set to $|T|$ periods.

Decision variables regard the structure of the network, i.e. the installation of production facilities of a certain technology and capacity $m$ at location $p$ in period $t$ ($x_{mpt}$). Additionally, allocation variables are determined, i.e. the biomass type $b$ delivered from source $q$ to module $m$ at location $p$ in period $t$ $\left(y_{bqmpt}^{biomass}\right)$, the slurry delivered from module $g$ at location $n$ to module $m$ at location $p$ in period $t$ ($y_{gnmpt}^{slurry}$), and the biodiesel transported from module $m$ at location $p$ to sink $s$ in period $t$ $\left(y_{mpst}^{biodiesel}\right)$. If production is lower than demand because of limited capacities, bio-diesel is purchased externally ($v_{st}$).

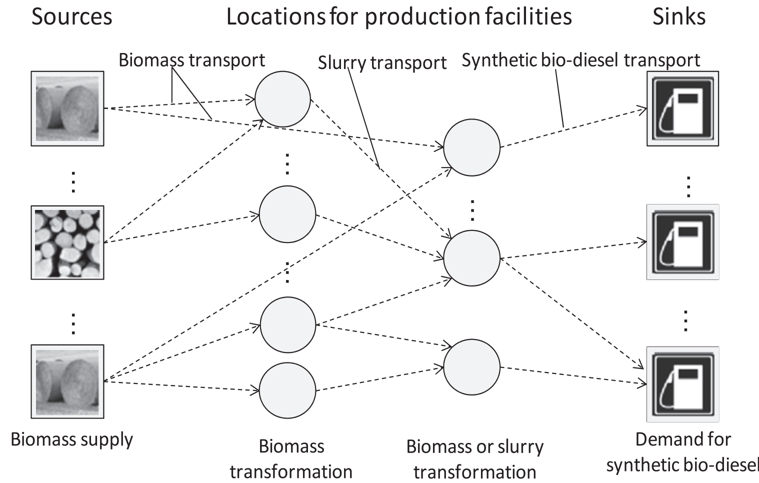
The following notation is used:

*Indices*

| | |
|---|---|
| $A$ | index set of scenarios; $a \in A$ |
| $B$ | index set of biomass types; $b \in B$ |
| $M$ | index set of modules; $m, g \in M; M = M_{in}^{biomass} \cup M_{in}^{slurry}$; $M = M_{out}^{slurry} \cup M_{out}^{biodiesel}$ |
| $M_{in}^{biomass}$ | index subset of modules processing biomass; $m \in M_{in}^{biomass}$ |
| $M_{in}^{slurry}$ | index subset of modules processing slurry; $m \in M_{in}^{slurry}$ |
| $M_{out}^{biodiesel}$ | index subset of modules producing synthetic bio-diesel; $m \in M_{out}^{biodiesel}$ |
| $M_{out}^{slurry}$ | index subset of pyrolysis module producing slurry; $g \in M_{out}^{slurry}$ |
| $P$ | index set of potential locations; $p, n \in P$ |
| $Q$ | index set of sources; $q \in Q$ |
| $S$ | index set of sinks; $s \in S$ |
| $T$ | index set of periods; $t \in T$ |

*Parameters and coefficients*

| | |
|---|---|
| $Biom_{bqt}$ | available biomass $b$ at source $q$ in period $t$ |
| $c_{bt}$ | costs for one mass unit of biomass $b$ in period $t$ |
| $\gamma_{bm}^{biodiesel}$ | transformation coefficient of module $m$ which describes the transformation of biomass $b$ into bio-diesel |
| $\gamma_{b}^{slurry}$ | transformation coefficient which describes the transformation of biomass $b$ into slurry |
| $\gamma_{m}^{biodiesel}$ | transformation coefficient of module $m$ which describes the transformation of slurry into bio-diesel |
| $\varepsilon_{b}^{biomass}$ | capacity unit coefficient, converting a mass unit of biomass $b$ into the capacity unit of modules |
| $\varepsilon^{slurry}$ | capacity unit coefficient, converting a mass unit of slurry into the capacity unit of modules |
| $maxcap_m$ | upper capacity limit of module $m$ |
| $Invest_m$ | investment costs for module $m$ |
| $Fix_m$ | fixed costs for operating module $m$ |
| $Proc_m$ | transformation costs of one capacity unit of input material (biomass or slurry) in module $m$ |
| $Demand_{st}$ | demand for synthetic bio-diesel in sink $s$ in period $t$ |
| $r_t$ | revenues owing to the sale of one mass unit of synthetic bio-diesel in period $t$ |
| $p_t$ | penalty costs for purchasing a mass unit of synthetic bio-diesel in period $t$ |
| $d_{qp}$ | distance between source $q$ and location $p$ |
| $d_{np}$ | distance between location $n$ and location $p$ |
| $d_{ps}$ | distance between location $p$ and sink $s$ |
| $c_{b}^{trans,biomass}$ | distance dependent transportation costs for one mass unit of biomass $b$ |
| $c_{b}^{handl,biomass}$ | fixed transportation costs for handling one mass unit of biomass $b$ |
| $c^{trans,slurry}$ | distance dependent transportation costs for one mass unit of slurry |




**Decision variables:**

| | | | |
|---|---|---|---|
| Network structure | Installation of production facilities *(technology/capacity, location, period)* $x_{mpt}$ | | |
| Material transport | Biomass *(source, delivering or purchasing production facility,* $y^{biomass}_{bqmpt}$ | Slurry *production facility,* $y^{slurry}_{gnmpt}$ | Synthetic bio-diesel *location, sink, period)* $y^{biodiesel}_{mpst}$ |
| External purchase | | | Synthetic bio-diesel *(sink, period)* $v_{st}$ |

**Fig. 3.1.** Production network for synthetic bio-diesel.

$c^{handl,slurry}$ fixed transportation costs for handling one mass unit of slurry
$c^{trans,biodiesel}$ distance dependent transportation costs for one mass unit of synthetic bio-diesel
$c^{handl,biodiesel}$ fixed transportation costs for handling one mass unit of synthetic bio-diesel
$I_t$ sum of investment costs in period $t$
$R_t^{material\ flow}$ sum of net revenues from material flows in period $t$
$C_t^{process}$ sum of costs for operating production processes in period $t$
$NPV$ net present value
$NPV_a$ net present value for scenario $a$
$NPV^{min}$ aspiration level
$o_a$ occurrence probability of scenario $a$
$i$ Interest rate

*Decision variables*

$x_{mpt}$ binary decision variable taking value 1 if module $m$ is opened in period $t$ at location $p$, else 0
$z_{mpt}$ number of modules $m$ at location $p$ which are already open during period $t$
$y^{biomass}_{bqmpt}$ biomass $b$ being transported from source $q$ to module $m$ at location $p$ in period $t$
$y^{slurry}_{gnmpt}$ mass of slurry being transported from module $g$ at location $n$ to module $m$ at location $p$ in period $t$
$y^{biodiesel}_{mpst}$ mass of synthetic bio-diesel being transported from module $m$ at location $p$ to sink $s$ in period $t$
$v_{st}$ compensation mass of synthetic bio-diesel purchased at sink $s$ in period $t$

We introduce the objective function of the basic model first, and then describe the constraints.

*3.1.1. Objective function*

Since we want to evaluate under which conditions the production of synthetic bio-diesel is beneficial, we present a profit oriented approach based on the net present value ($NPV$) as result of discounted net revenues of material flows $R_t^{material\ flow}$ minus discounted investment costs $I_t$ minus discounted costs for operating production processes $C_t^{process}$.

$$Max\ NPV = \sum_{t \in T} \left[ R_t^{material\ flow} - I_t - C_t^{process} \right] \cdot (1+i)^{-t} \tag{1}$$

Total revenues for material flows result from the sale and purchase of materials and from transports.

Revenues can be achieved by selling bio-diesel $\left(y^{biodiesel}_{mpst}\right)$. The costs for buying biomass $\left(y^{biomass}_{bqmpt}\right)$ depend on amount and consistency. Additionally, penalty costs occur for external purchase of bio-diesel ($v_{st}$.) if demand cannot be satisfied by the network.

Transportation costs consist of a distance dependent part and a part for materials handling. Transports of biomass take place between sources and production facilities, transports of slurry between production facilities (in case of decentralized plant concepts), and transports of synthetic bio-diesel between production facilities and sinks.

$$\begin{aligned} R_t^{material\ flow} = & \sum_{m \in M_{out}^{biodiesel}} \sum_{p \in P} \sum_{s \in S} y^{biodiesel}_{mpst} \cdot r_t \\ & - \sum_{b \in B} \sum_{q \in Q} \sum_{m \in M_{in}^{biomass}} \sum_{p \in P} y^{biomass}_{bqmpt} \cdot c_{bt} - \sum_{s \in S} v_{st} \cdot p_t \\ & - \left[ \sum_{b \in B} \sum_{q \in Q} \sum_{m \in M_{in}^{biomass}} \sum_{p \in P} y^{biomass}_{bqmpt} \cdot \left( c_b^{trans,biomass} \cdot d_{qp} + c_b^{handl,biomass} \right) \right. \\ & + \sum_{g \in M_{out}^{slurry}} \sum_{m \in M_{in}^{slurry}} \sum_{n,p \in P} y^{slurry}_{gnmpt} \cdot \left( c^{trans,slurry} \cdot d_{np} + c^{handl,slurry} \right) \\ & \left. + \sum_{m \in M_{out}^{biodiesel}} \sum_{p \in P} \sum_{s \in S} y^{biodiesel}_{mpst} \cdot \left( c^{trans,biodiesel} \cdot d_{ps} + c^{handl,biodiesel} \right) \right] \\ & \forall t \in T \end{aligned} \tag{2}$$

Investment costs result from the installation of a module $m$ at location $p$ in period $t$. Due to the different considered capacities of the modules and the involved economies of scale, only one module is to be installed at a location in each planning period. Since an increasing demand for synthetic bio-diesel is assumed, de-installation of production modules is not allowed.

$$I_t = \sum_{m \in M} \sum_{p \in P} Invest_m \cdot x_{mpt} \quad \forall\ t \in T. \tag{3}$$

Costs for operating production processes consist of a fixed part and a capacity utilization depending part.

$$C_t^{process} = \sum_{m \in M} \sum_{p \in P} \Bigg[ Fix_m \cdot z_{mpt} + Proc_m \cdot \Bigg( \sum_{b \in B} \sum_{q \in Q} y_{bqmpt}^{biomass} \cdot \varepsilon_b^{biomass} + \sum_{g \in M_{out}^{slurry}} \sum_{n \in P} y_{gnmpt}^{slurry} \cdot \varepsilon^{slurry} \Bigg) \Bigg] \quad \forall\ t \in T. \tag{4}$$

### 3.1.2. Mass balances

Modules can be divided into modules $g$ producing slurry $\left(g \in M_{out}^{slurry}\right)$ and modules $m$ producing synthetic bio-diesel $\left(m \in M_{out}^{biodiesel}\right)$. Within decentralized plant concepts, biomass is transformed into slurry at location $n$. This transformation process is described using the transformation coefficient $\gamma_b^{slurry}$. The produced slurry is then transported to module $m$ at location $p$ for further processing.

$$\sum_{b \in B} \sum_{q \in Q} y_{bqgnt}^{biomass} \cdot \gamma_b^{slurry} = \sum_{m \in M_{in}^{slurry}} \sum_{p \in P} y_{gnmpt}^{slurry} \quad \forall\ g \in M_{out}^{slurry}; \quad n \in P; \quad t \in T. \tag{5}$$

Biomass and slurry can be used for production of synthetic bio-diesel in module $m$ $\left(m \in M_{out}^{biodiesel}\right)$. Again, transformation coefficients describe the transformation process of biomass $\left(\gamma_{bm}^{biodiesel}\right)$ respectively slurry $\left(\gamma_m^{biodiesel}\right)$ into bio-diesel. The bio-diesel is then delivered to the sink $s$ in order to satisfy demand.

$$\sum_{b \in B} \sum_{q \in Q} y_{bqmpt}^{biomass} \cdot \gamma_{bm}^{biodiesel} + \sum_{g \in M_{out}^{slurry}} \sum_{n \in P} y_{gnmpt}^{slurry} \cdot \gamma_m^{biodiesel} = \sum_{s \in S} y_{mpst}^{biodiesel} \quad \forall\ m \in M_{out}^{biodiesel}; \quad p \in P; \quad t \in T. \tag{6}$$

### 3.1.3. Capacity restrictions

Constraints 7 ensure that biomass of type $b$ taken from source $q$ in period $t$ does not exceed the available amount of biomass at this location.

$$\sum_{m \in M_{in}^{biomass}} \sum_{p \in P} y_{bqmpt}^{biomass} \leqslant Biom_{bqt} \quad \forall\ b \in B; \quad q \in Q; \quad t \in T. \tag{7}$$

Production modules are restricted by an upper capacity limit $maxcap_m$ (constraints 8). The actual available capacity at a site is constrained by the number of open modules $z_{mpt}$, i.e. the sum of modules that were already installed until period $t$. Depending on the capacity units of modules, a conversion of input material masses (biomass respectively slurry) is necessary. This conversion is described using a specific capacity unit coefficient for biomass $\left(\varepsilon_b^{biomass}\right)$ respectively for slurry $\varepsilon^{slurry}$.

$$\sum_{b \in B} \sum_{q \in Q} y_{bqmpt}^{biomass} \cdot \varepsilon_b^{biomass} + \sum_{g \in M_{out}^{slurry}} \sum_{n \in P} y_{gnmpt}^{slurry} \cdot \varepsilon^{slurry} \leqslant maxcap_m \cdot z_{mpt} \quad \forall\ m \in M; \quad p \in P; \quad t \in T. \tag{8}$$

The demand in sinks $s$ causes both, the installation of production capacities and the flow of materials within the network. This demand is satisfied by bio-diesel $y_{mpst}^{biodiesel}$, which is produced in the network. Additionally, unsatisfied demand can be compensated by externally purchased bio-diesel $v_{st}$.

$$Demand_{st} = \sum_{m \in M_{out}^{biodiesel}} \sum_{p \in P} y_{mpst}^{biodiesel} + v_{st} \quad \forall\ s \in S; \quad t \in T. \tag{9}$$

### 3.1.4. Variable declaration

Binary variables represent the decision as to whether a specific module $m$ is opened at location $p$ in period $t$ or not.

$$x_{mpt} \in \{0, 1\} \quad \forall m \in M; \quad p \in P; \quad t \in T \tag{10}$$

The number of modules $m$ that were already opened at location $p$ until period t is described by counting variables.

$$z_{mpt} = \begin{cases} x_{mpt}, & for\ t = 1 \\ z_{mpt-1} + x_{mpt}, & else \end{cases} \quad \forall\ m \in M; \quad p \in P; \quad t \in T \tag{11}$$

Within the network, material flows can only take on positive values.

$$y_{bqmpt}^{biomass} \geqslant 0 \quad \forall\ b \in B; \quad q \in Q; \quad m \in M_{in}^{biomass}; \quad p \in P; \quad t \in T, \tag{12}$$

$$y_{gnmpt}^{slurry} \geqslant 0 \quad \forall\ g \in M_{out}^{slurry}; \quad m \in M_{in}^{slurry}; \quad n, p \in P; \quad t \in T, \tag{13}$$

$$y_{mpst}^{biodiesel} \geqslant 0 \quad \forall; \quad m \in M_{out}^{biodiesel}; \quad p \in P; \quad s \in S; \quad t \in T. \tag{14}$$

## 3.2. Scenario based extension

Existing uncertainties lead to risks for the decision makers. The risk attitude of decision makers depends on their individual conditions. In order to account for diverse risk attitudes, the basic model is extended by applying different objective functions. Network configurations are evaluated based on resulting net present values of single scenarios $a$ ($NPV_a$), which consist of discounted revenues and costs of scenario specific material flows, investments and process operation. Four different decision criteria are selected as objective functions (cp. Table 3.1) that represent a wide range of risk attitudes. This allows for a differentiated analysis and assessment of results.

The maxmin criterion (I) represents extremely risk averse decision makers. The criterion focuses on the worst development solely, which is to be optimized. Subjective occurrence probabilities of scenarios $o_a$ are not required. Criterions (II)–(IV), however, represent a broader range of risk attitudes. The expected-value criterion is often quoted to be risk neutral, since both negative and positive variations are considered equally. However, the represented risk attitude cannot be assigned, as scenario specific results may vary strongly from expected results. We select the expected-value criterion, since it is established in practice and is easily comprehensible. The Hodges–Lehmann criterion (III) considers not only the expected value, but also worst case developments. Thus, a more risk averse attitude is represented. If worst case developments can be predicted in advance, the Hodges–Lehmann criterion leads to solutions which are relatively stable against negative future developments (Scholl, 2001). When applying the expected value-expected failure criterion, an aspiration level ($NPV^{min}$) has to be defined in advance. Depending on how the weighting factor is set, negative variations from the aspiration level are penalized. This criterion promises stable solutions near the aspiration level for most cases. However, opportunities of possible positive developments are neglected.

For the Hodges–Lehmann criterion or the expected-value-expected-failure criterion, the setting of the weighting factors influences the represented risk attitude of decision makers. Therefore, the decision makers have to be involved in setting these parameters. Analyzing the sensitivity of the resulting network structures

**Table 3.1**
Objective functions.

| | Criterion | Risk attitudes | Objective function |
|---|---|---|---|
| I | Maxmin | Extreme risk averse | $Max\ min\{NPV_a \mid a \in A\}$ |
| II | Expected-value | | $Max\ E(NPV) = \sum_{a \in A} o_a \cdot NPV_a$ |
| III | Hodges–Lehmann | Dependent on weighting factor $\mu$ | $Max\ \mu \cdot E(NPV) + (1-\mu) \cdot min\ \{NPV_a \mid a \in A\}\ with\ 0 \leqslant \mu \leqslant 1$ |
| IV | Expected-value-expected-failure | Dependent on a factor $\omega$ | $Max\ E(NPV) - \omega \cdot \sum_{a \in A} o_a \cdot (max\{0, NPV^{min} - NPV_a\})$ |

to these weighting factors can increase transparency of the decision making situation.

The constraints of the basic model have to be adapted for the scenario based approach. Adoptions refer to control variables (material flows in the network) as well as to scenario specific data (biomass potential, investment related costs, demand for bio-diesel).

## 4. Case study

Research studies show that regions with a high biomass potential should be chosen for the installation of bio-diesel production plants. That way, transports of bulky raw material can be reduced (Leduc et al., 2010; Kerdoncuff, 2008). Hence, in the case study we will focus on the federal state of Niedersachsen, which has the second highest biomass potential in Germany (Niedersächsisches Ministerium für Ernährung, 2009). Additionally, we consider the federal states Bremen and Hamburg. Both states are directly adjacent to Niedersachsen and are indicated by a relatively high diesel demand due to a high population density.

### *4.1. Data*

Investment decisions can only be evaluated if all relevant data is known. Therefore, the relevant data regarding biomass supply, production technologies and demand for synthetic bio-diesel is presented in the following. For the case study, a planning horizon of 20 years (starting in 2008) is used. The interest rate is set to 8% per year.

#### *4.1.1. Supply of biomass*

The biomass provision takes place in gathering points, which are supplied by local farmers and lumberjacks. Here, different types of biomass are collected, such as energy crops and residual materials like straw and logging remains. These materials differ with regard to moisture and energy content.

In order to forecast future biomass availability, two scenarios were developed in the European research project "Renew", (cp. Table 4.1; Renew, 2004).

One scenario assumes an ecological cultivation avoiding negative environmental effects, while a maximization of area yields is aimed at in the second scenario. Both scenarios assume that food and fibre production are not influenced, agricultural yields increase over time, and set-aside land is used for the cultivation of energy crops. Based on these forecasts, biomass specific yearly yield factors per land unit are determined for 2020 for Niedersachsen, Hamburg and Bremen (Table 4.1).

The biomass potential is then calculated based on yield factors and information on land use as shown in Fig. 4.1 for the example of straw. The prices for biomasses are calculated for an average collection distance of 30 km based on data of (Renew, 2004; Suurs, 2002).

#### *4.1.2. Production Technologies for synthetic bio-diesel*

Since the technology concepts of Choren (2011) and of For-schungszentrum Karlsruhe (FZK) (Henrich et al., 2007) are furthest evolved, these two concepts are considered within the case study. A Choren plant with a capacity of 15,000 Mg diesel/*a* is already installed in Freiberg since 2008. FZK operates a pilot plant in Karlsruhe.

*4.1.2.1. Capacities.* Synthesis modules and the centralized plant concept are considered in two capacity classes, 500 MW and 2 GW. Thus, economies of scale are taken into account. The pyrolysis module of the FZK has a capacity of 100 MW. The economies of scale concern both the required investment costs for the production plants and the costs for production processes. In addition to the different capacity classes, there is a choice of two different synthesis methods: Fischer–Tropsch-Synthesis (FTS) and Methanol Synthesis (MS). Transformation coefficients are determined according to mass balances of Beiermann (2011).

Costs for operating production processes are divided into fixed costs and costs that depend on the capacity utilization. The fixed costs cover costs for insurances, maintenance and operating supply. Costs that depend on capacity utilization are caused by energy and material flows that are determined based on the flow sheeting simulations of Beiermann (2011). Costs for operational staff, administration, and operating supervision are determined based on a concave function of capacity utilization according to Hamelinck et al. (2004). Initial values are costs for operating staff of a 400 MW plant, the scaling exponent is 0.25, and 8000 operating hours are assumed per annum.

Investment costs for technologies are determined in a two-step procedure. In the first step, single investment costs of the main equipments are determined. Current investment costs can be estimated based on past data of investment costs for similar equipments of different capacities using scale and time indices like the Chemical Engineering Plant Cost Index (Peters and Timmerhaus, 2002; Beiermann, 2011). The equipment specific scale exponents

**Table 4.1**
Yield factors respectively yields (calculated on the basis of Renew (2004)).

| Type of biomass | Supplying state | Initial value | 2020: High area yield | 2020: Ecological cultivation |
|---|---|---|---|---|
| Logging residues | Germany | 48.50 (Mg/(km$^2$ a)) | 53.31 (Mg/(km$^2$ a)) | 52.55 (Mg/(km$^2$ a)) |
| Straw | Hamburg | 0 | 0 | 0 |
| | Bremen | 0 | 0 | 0 |
| | Niedersachsen | 44.70 (Mg/(km$^2$ a)) | 33.64 (Mg/(km$^2$ a)) | 25.88 (Mg/(km$^2$ a)) |
| Fast-growing lumber | Hamburg | 4787.23 (Mg/a) | 12234.04 (Mg/a) | 5319.15 (Mg/a) |
| | Bremen | 13.81 (Mg/(km$^2$ a)) | 64.45 (Mg/(km$^2$ a)) | 13.81 (Mg/(km$^2$ a)) |
| | Niedersachsen | 44.55 (Mg/(km$^2$ a)) | 157.15 (Mg/(km$^2$ a)) | 66.41 (Mg/(km$^2$ a)) |

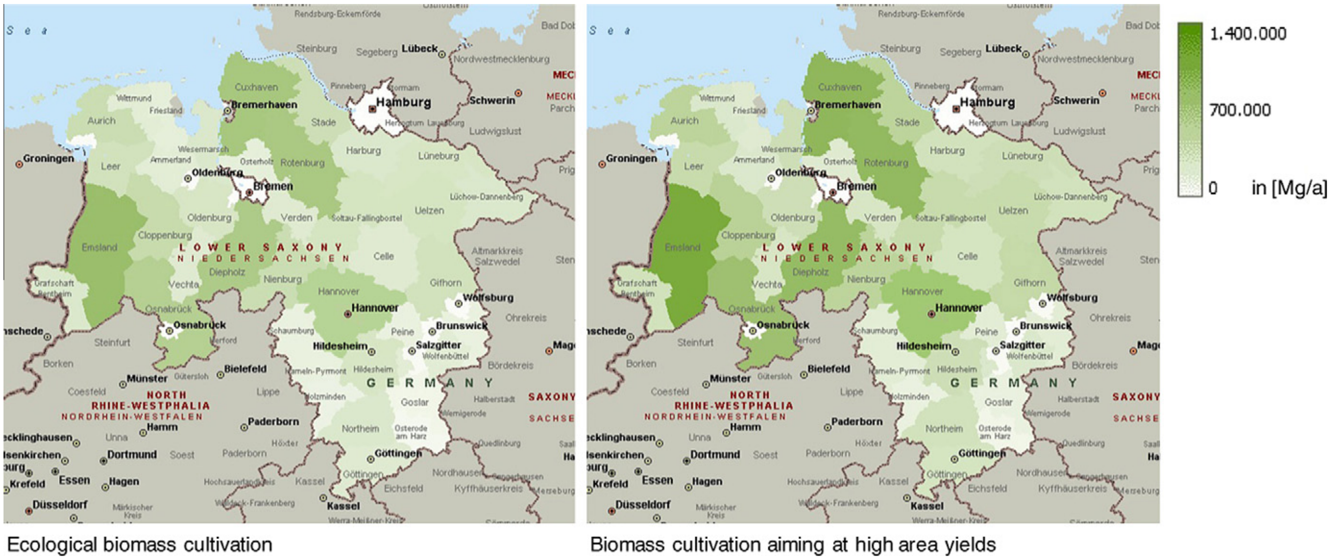



**Fig. 4.1.** Technically utilizable straw potential in the year 2020 for different scenarios.

range between 0.6 and 1 (Peters and Timmerhaus, 2002; Hamelinck et al., 2004). In the second step, we consider auxiliary investment costs applying surcharge factors to the sum of the main equipments' investment costs. According to (Peters and Timmerhaus, 2002) surcharge factors amount to 3.74 for green field development. The resulting modules with investment and process costs are presented in Table 4.2.

### 4.1.3. Demand for synthetic bio-diesel

The demand for synthetic bio-diesel is determined by legal regulations and market effects. The latest EU directive 2009/28/EC requires a 10% substitution of fossil fuel through bio-fuel by the year 2020. The directive is implemented by the German bio-fuel quota law setting annual quotas and fuel type specific quotas (cp. Fig. 4.2). However, former revisions of directives on European and national level have shown that the quotas are uncertain and changes may occur in the long-term.

In order to calculate the spatial distribution of the overall diesel demand in the considered region, we first apply a polynomial approximation to calculate the overall diesel demand based on the forecast of the German Oil Industry Association (cp. Fig. 4.3; MWV, 2009). Then, a Germany-wide dynamic per capita diesel demand is deduced, combining the approximated diesel demand with the forecasted population development as given by German Federal Statistical Office (2006). Eventually, the demand for synthetic bio-diesel is determined according to legal regulations on bio-fuel quotas. Future prices for diesel are adjusted by linear extrapolation based on data of German Federal Statistical Office (2009).

## 4.2. Scenarios

Currently, neither political decision makers nor potential investors can foresee, how conditions for the production of synthetic bio-diesel will develop in the future. Thus, possible future developments are described by scenarios.

Within this paper, we start with a base case and generate other scenarios combining selected values of the uncertain parameters in the fields of biomass potential, investment cost for production technologies and demand for synthetic bio-diesel:

- Regarding future biomass potential, two possible developments are considered (Renew, 2004): an ecological cultivation avoiding negative environmental effects, and a cultivation aiming at a maximization of area yields.
- With regard to production technologies, we focus on uncertainties of investment costs. In literature, estimates of these values differ strongly (cp. e.g. Beiermann, 2011; Kerdoncuff, 2008; Hamelinck et al., 2004). We lay studies of Beiermann (2011) as basis and additionally consider 5% lower investment costs, according to the estimation method of Peters and Timmerhaus (2002).
- The uncertain demand for synthetic bio-diesel is described by three different developments, regarding the quota on overall diesel demand. One development corresponds to current legal regulations on the quota of bio-fuels on total fuels sales, a second development describes a lower quota according to minimum quota of bio-diesel, and the third development follows former EU targets of 20% substitution by 2020.

**Table 4.2**
Production technologies.

| Plant concept | Module | Capacity | Synthesis | Investment (€) | Annual fixed costs (€) | Capacity utilization dependent costs (€/kW) |
|---|---|---|---|---|---|---|
| Choren | 1 | 500 MW | FTS | 1,088,915,578 | 79,136,940 | 33.70 |
| | 2 | 500 MW | MS | 1,206,828,521 | 87,706,263 | 52.32 |
| | 3 | 2 GW | FTS | 3,299,471,987 | 239,789,127 | 31.06 |
| FZK | 4 | 500 MW | FTS | 691,701,939 | 50,269,438 | −8.44 |
| | 5 | 500 MW | MS | 817,164,392 | 59,387,422 | 27.75 |
| | 6 | 2 GW | FTS | 2,144,481,524 | 155,850,195 | −11.08 |
| | 7 | 2 GW | MS | 2,370,736,570 | 172,293,280 | 25.11 |
| FZK pyrolysis | 8 | 100 MW | | 57,145,024 | 4,153,015 | 32.87 |

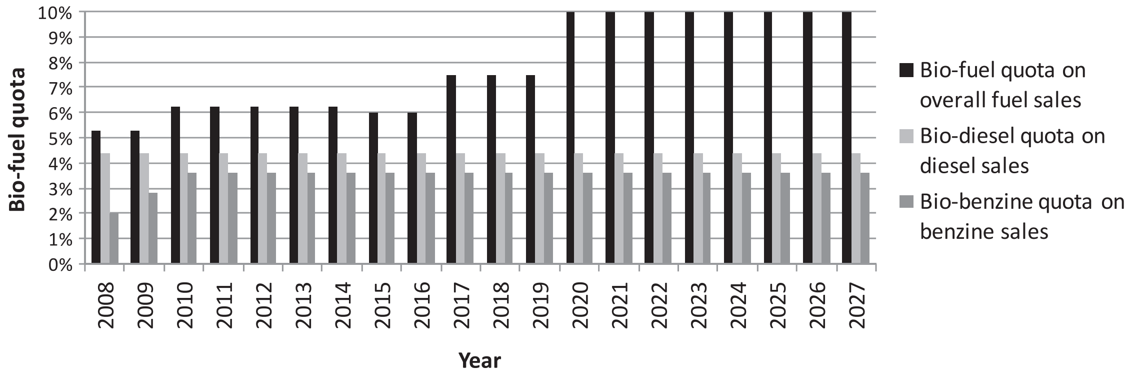



**Fig. 4.2.** Mandatory bio-fuel quota according to the German bio-fuel quota law.

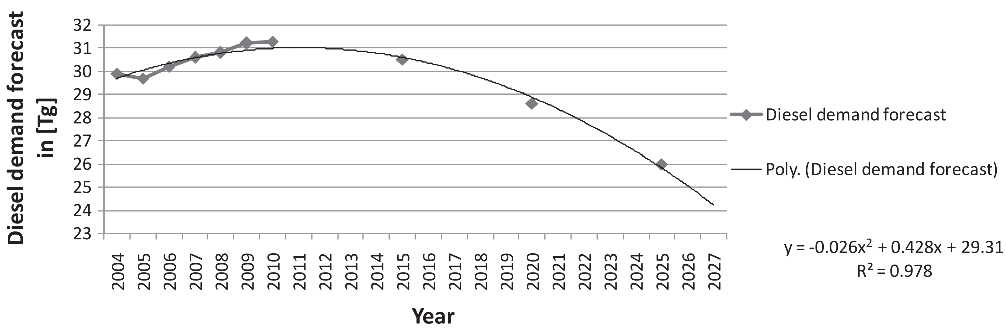



**Fig. 4.3.** Diesel demand forecast (own calculations based on data from MWV (2009)).

In the base scenario, ecological biomass cultivation, investment costs for production technologies according to Beiermann (2011), as well as demand according to current legal regulations on the minimum quota of bio-fuel are assumed. Based on the basic scenario, uncertain parameters are varied one by one. Resulting scenarios are summarized in Table 4.3.

### *4.3. Implementation*

For the basic model, a pre-processing is carried out in order to reduce model complexity. Sites for centralized plant concepts and for synthesis modules of decentralized plant concepts are pre-selected using biomass availability, connection to infrastructure, and marketing opportunities as site related factors (Panichelli and Gnansounou, 2008). Ten sites are selected for the centralized concept as well as for the synthesis device of the decentralized concept. The basic model was implemented in ILOG OPL (version 6.1.1) and calculated on a 64 bit Windows machine with a working memory of 12 GB RAM. The resulting model size amounts to 11,865 constraints, 2,380 binary and 2,380 integer decision variables and 729,701 non-zero-elements.

For the scenario based approach, we conducted a further pre-processing based on the results of the basic model regarding advantageous plant concepts. After this further pre-processing, resulting models amount to approximately 21,610 constrains, 600 binary and 600 integer decision variables and 1,932,180 non-zero-elements.

In all calculations, the permitted MIP gap was limited to 1%. Tests with different smaller problem instances of our planning problem showed that resulting deviations can be neglected considering the multiple uncertainties of this long-term planning decision. This is also affirmed by literature, e.g. Cordeau et al. (2006).

### *4.4. Resulting network configurations*

Network configurations that consider the scenarios separately are presented first, followed by network configurations for different risk attitudes of decision makers. Thus, we gather an understanding of influencing factors.

#### *4.4.1. Single scenario analysis*

Network configurations for single scenarios are valuated regarding their net present value. The resulting network

**Table 4.3**
Parameter values of considered scenarios.

| Parameter | Expression | Base scenario | High biomass potential | Low in-vestment costs | Low demand | High demand |
|---|---|---|---|---|---|---|
| Biomass provision | Ecological cultivation | X | | X | X | X |
| | High biomass yields | | X | | | |
| Techno-logies | Investment costs according to Beiermann (2011) | X | X | | X | X |
| | Low investment costs | | | X | | |
| Demand | Low quota | | | | X | |
| | Base quota | X | X | X | | |
| | High quota | | | | | X |

configurations at the end of the planning horizon are represented in Fig. 4.4. Regarding the geographical distribution of the installed production facilities it can be seen that facilities are concentrated in the administrative districts Diepholz, Nienburg, Rotenburg, and the Region Hannover for all scenarios. These administrative districts are characterized by a high biomass potential and have a relatively central position from a distribution point of view. The administrative districts Diepholz, Nienburg, and the Region Hannover particularly stand out due to a good position in terms of biomass provision. Regarding Rotenburg, the position between cities with high fuel demand (Hamburg and Bremen) is advantageous.

The solutions have in common that only centralized plant concepts are selected, and that modules are all installed in the first planning period. In all but the "low demand" scenario, production facilities are equipped with the Fischer–Tropsch-Synthesis. The Fischer–Tropsch-Synthesis has less process yields in comparison to the Methanol Synthesis, but is in turn characterized by lower investment costs. As the considered region is indicated by a relatively high biomass potential, transport distances are short and the characteristics of Fischer–Tropsch-Synthesis are advantageous. The installation of a centralized plant with the Methanol Synthesis in the "low demand" scenario is caused by both the demand, which cannot be satisfied by one 500 MW plant, and the site Rotenburg, which is in case of ecological biomass cultivation less advantageous in terms of the biomass purchase. Both factors make the higher process yields beneficial. Depending on demand for synthetic bio-diesel, plants are built with a capacity of either 500 MW or 2 GW.

These results reveal that the demand for synthetic bio-diesel has an essential influence on the chosen plant concepts and capacities. Biomass potential affects the sites of production facilities and the synthesis process if appropriate.

In Fig. 4.5 an overview of the proportion of discounted costs is given. In all scenarios, penalty costs occur, i.e. synthetic bio-diesel is purchased externally, as it is only profitable to operate plants with a high degree of capacity utilization. In all scenarios, the discounted transportation costs amount to low percentages only. The reason is that sourcing can be done from nearby places due to the high biomass potential in the considered region. Costs for biomass range from 17% to 21% of total costs. Investment costs for production technologies as well as fixed costs for operating production processes constitute a high amount of the resulting total discarded costs. On the one hand, this is due to the fact that current price indexes used for calculation of investment costs are relatively high. On the other hand, the planning horizon is limited to 20 years, and thus utilization times of plants are assumed to be 20 years. If utilization times of the production facilities exceed the planning horizon, the proportionate composition of the discounted cash flows will change.

All discounted costs within the planning horizon of 20 periods are represented in Fig. 4.6. Particularly, an extension of the utilization time will further increase the achievable NPV.

Fluctuations of discounted periodic costs result because of fluctuations in demand. Additionally, penalty costs for purchasing synthetic bio-diesel from outside the network occur. In Fig. 4.7, the discounted revenues for selling the synthetic bio-diesel as well as

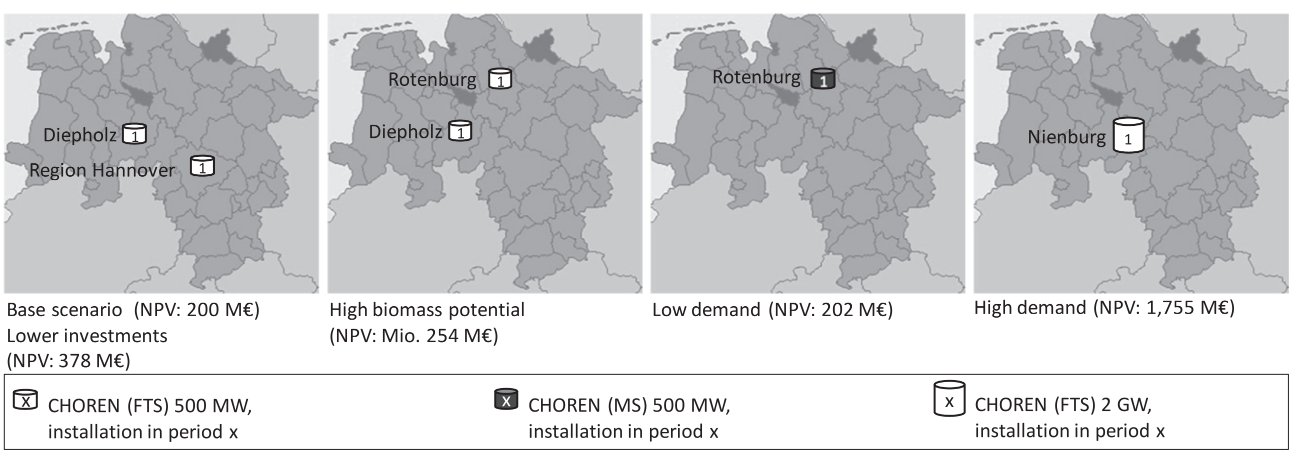



**Fig. 4.4.** Network configurations for single scenarios.

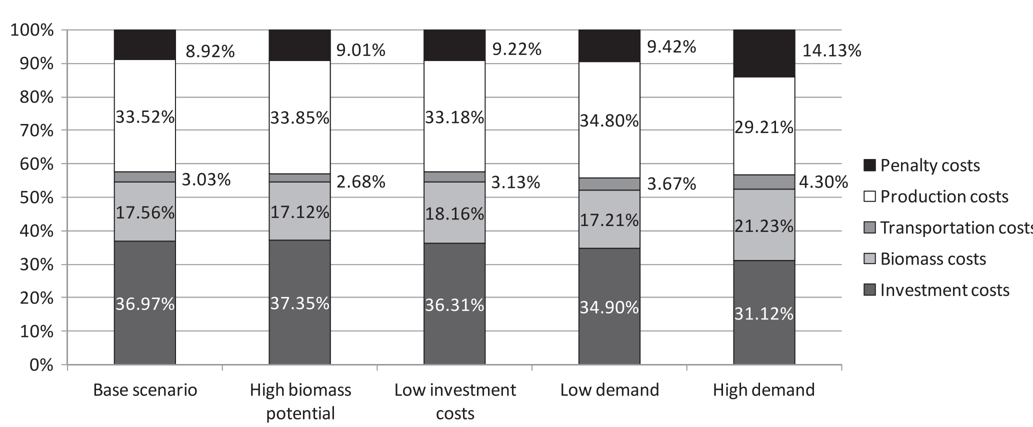



**Fig. 4.5.** Proportions of the discounted costs and investment costs for a planning horizon of 20 years.

the discounted costs for purchase of bio-diesel are exemplarily shown for the base scenario.

Due to the result that centralized plant concepts are advantageous in the considered region in Northern Germany, we further reduce the problem instance to only centralized plant concepts. In addition, we conduct a warm start using a solution of the single scenario analysis in order to reduce computing times: opening a centralized plant with a capacity of 2 GW in the first planning period in the administrative district Nienburg.

*4.4.2. Results for different risk attitudes of decision makers*

The network configurations for different risk attitudes of decision makers are presented in Fig. 4.8. Configurations are comparable to the single scenario analysis. The Fischer–Tropsch-Synthesis is used in the centralized plant concepts. However, the different risk attitudes of decision makers influence the chosen capacity of plants as well as the installation period.

The maxmin criterion and the Hodges–Lehmann criterion result in an installation of a 500 MW module in Nienburg in the first planning period, and a 2 GW module in Rotenburg in the ninth planning period. This network constellation represents a strong risk averse attitude of decision makers. By installing only one 500 MW module at the beginning of the planning horizon, investments are comparatively low and a high flexibility is provided regarding a future capacity adjustment. Depending on how the demand for synthetic bio-diesel will develop, further modules of different capacity can be installed. However, only a small portion of demand could be satisfied if high demand for synthetic bio-diesel arises in the first planning periods. Thus, market potentials would be wasted.

Using the expected-value-expected-failure criterion leads to an installation of two 500 MW modules in the first planning period in the regions Diepholz and Hannover. A third 500 MW module is installed in the ninth planning period in Rotenburg. Thus, a higher readiness to take risks is represented compared to the previous network constellation. Installing two 500 MW modules in the first planning period increases the risk of excess capacity. However, if demand for synthetic bio-diesel develops positive, a higher demand can be met. Moreover, this network configuration still provides flexibility regarding further capacity installation.

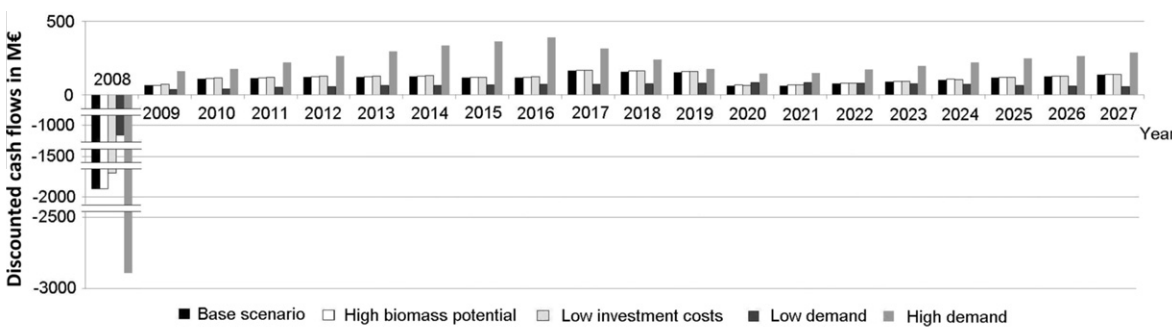



**Fig. 4.6.** Discounted periodical cash flows within the planning horizon.

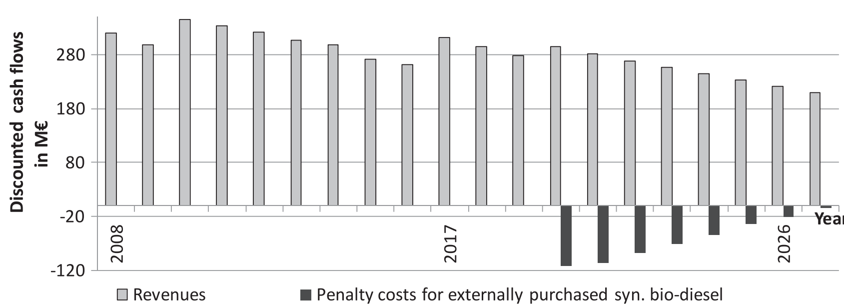



**Fig. 4.7.** Discounted revenues and penalty cost for base scenario.

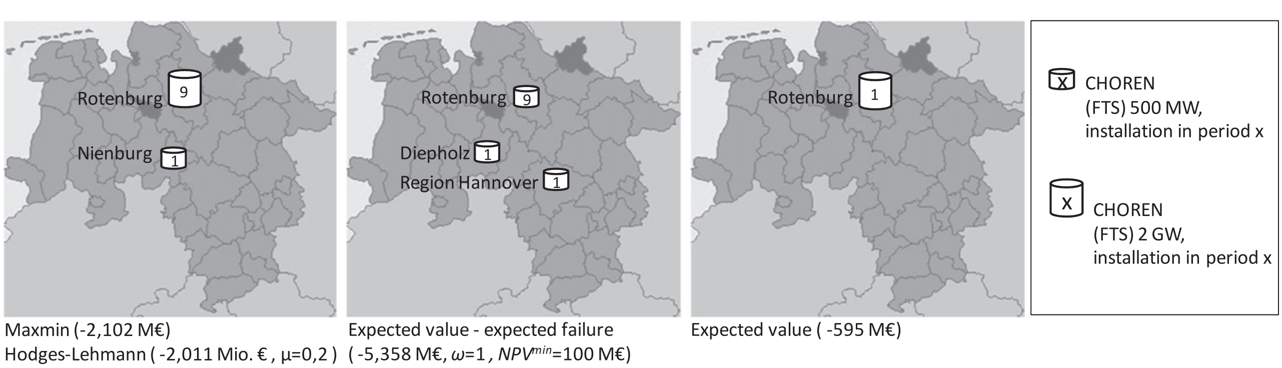



**Fig. 4.8.** Network configurations of scenario based approach.

The installation of a 2 GW module in the first planning period in Rotenburg results for the expected-value criterion. This network configuration represents the highest readiness to take risks by the decision makers. By installing a 2 GW module, the risk of excess capacity is high. The network configuration is advantageous only, if demand for synthetic bio-diesel develops according to the high demand scenario. Else, excess capacities occur. This network constellation provides economies of scale if demand develops very positive.

Based on these results, recommendations for political decision makers and potential investors are derived in the following.

### 4.5. Recommendations

Results of the case study clearly show that the uncertain future developments are a challenge for the design of regional production networks for synthetic bio-diesel. Optimal network configurations strongly depend on future demand. Moreover, a certain synthetic bio-diesel demand is required in order to make the investments in and the operation of production facilities worthwhile for investors. Hence, political decision makers shall ensure legal regulations and conditions in the long-term. Particularly, the mandatory bio-fuel quota should be fixed, and subsidies in terms of tax reliefs or tax exemptions should be assured in the long-term. Previous modifications in taxation for first generation bio-diesel have already caused shutdowns of production facilities.

With regard to potential investors, three essential recommendations are derived. Firstly, centralized plant concepts seem advantageous in regions characterized by a high biomass potential as they are a good compromise between the use of economies of scale and the reduction of biomass transports. Furthermore, small capacities provide the possibility to adapt production capacities to changing conditions. The selection of the synthesis process depends on the biomass potential of the region and on technological developments. In most cases, the Fischer–Tropsch-Synthesis proves to be beneficial. The use of the Methanol Synthesis promises higher diesel recoveries, but requires higher investments.

Secondly, a gradual installation of production capacities over time can be recommended. In principle, sites that are part of network configurations for different scenarios are advantageous for the installation of first plants. By conducting a continuous planning approach, one can gradually decide on the installation of further production facilities based on existing network structures. Doing so, new information can be taken into account and difficulties that result out of the limited planning horizon can be met. That way, an adaptation to changing conditions is possible and risks can be further reduced. Additionally, it would be easier to meet budget constraints that may arise if single decision makers invest in such a network.

Thirdly, we can state that transportation costs of biomass and bio-diesel are crucial for location planning. Within the conducted case study, the chosen sites turned out to be very similar with regard to transportation costs. In such cases, investors should consider further aspects in decision making, for instance long-term contracts with biomass suppliers.

## 5. Conclusion

A case study for integrated location, capacity and technology planning for production of synthetic bio-diesel is carried out based on current data. Thereby, centralized and decentralized plant concepts of different capacities are regarded. Uncertain developments for production of bio-diesel in the future are accounted for within scenarios. Advantageous technology concepts are determined regarding different risk attitudes of decision makers.

Since previous planning approaches for the design of networks for production of bio-fuels were aligned to either centralized or decentralized plant concepts only, the developed planning approach clearly represents an added value for the decision makers. Also, the developed scenario-based planning approach allows consideration of different risk attitudes of decision makers and of uncertainties in a proactive way.

Several prospects arise with regard to future work. So far, the calculations aim at an evaluation of the profitability of production networks in general. If a single decision maker is planning an investment, further restrictions like budget constraints would have to be regarded. The planning approach can also be applied to other case studies. For instance, bio-fuel production networks could be analyzed on an European and global scale regarding synergy effects with oil refineries and global biomass availability. If global sourcing networks for biomass are regarded, special attention has to be given to sustainability of these production networks, since social and ecological consequences in the developing countries have to be considered and managed. Also, uncertainty related to major disasters that might create disruptions in the network structure is neglected so far and might deserve special attention in future research. Due to the complexity of the models, specific solution procedures will have to be developed.

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